

IMPLEMENTATION PLAN & ROADMAP: THE LIVING MAP

Team [ANONYMIZED] — Phase 1 Submission

Independent Submission • IEEE RAS × IEEE AESS Tunisia Section TSY14 Technical Challenge

IMPLEMENTATION PLAN: THE LIVING MAP

Team [ANONYMIZED] — Phase 1 Submission Independent Submission Challenge: IEEE RAS × IEEE AESS Tunisia Section TSY14 Technical Challenge Environment: Mines / Tunnels | Final Deadline: 01/12/2026

1. PROJECT OVERVIEW & WORKSTREAMS

The Living Map project delivers spatial memory for emergency robots operating in GPS-denied, communication-denied subterranean tunnels. Work is organized across five parallel workstreams: 1. **WS1: Simulation & World Modeling** (Webots tunnel world, physics, gas dispersion, obstacle models) 2. **WS2: Writer Robot Subsystem** (LiDAR SLAM, frontier-based exploration, gas/blockage event detection, dispenser) 3. **WS3: Beacon Subsystem** (16-byte protocol codec, 433 MHz RF emulation, gossip relay, confidence aging decay) 4. **WS4: Outside Network Area (ONA) & Command Post** (RF gateway, edge translation, store-and-forward queue, live GPS map) 5. **WS5: Executor Robot Subsystem** (Mission briefing protocol, chain-following navigation, field verification, map refresh)

2. MILESTONE TIMELINE (ENDING 01/12/2026)

Phase	Window	Milestone	Key Deliverables	Dependencies	Risk & Mitigation
Phase 1	14/09 – 05/10/2026	System Architecture & Phase 1 Deliverables	Technical report (≤ 6 pages), system architecture (D1–D8), Webots world scaffold, repository structure, anonymity verification	Challenge spec	Scope creep: strict focus on Phase 1 rubric deliverables
Phase 2-A	06/10 – 19/10/2026	Writer Autonomy & Event Sensing	Occupancy grid mapping (5 cm cells), A* frontier exploration, virtual gas leak detector, blockage detection	Phase 1 architecture	Exploration stuck in dead ends: fallback to wall-following
Phase 2-B	20/10 – 02/11/2026	Beacon Protocol & ONA Gateway	Beacon PROTO, 16-byte codec, gossip relay (Emitter/Receiver), confidence aging, ONA edge processor, geodetic frame translation	Phase 2-A	RF collisions: random jitter (0–500 ms) and ALOHA backoff
Phase 2-C	03/11 – 16/11/2026	Executor Navigation & Command Post Live Map	Executor briefing protocol (ch3), beacon chain-following FSM, live map GPS rendering, batch testing (N=20 runs)	Phase 2-B	Command post sync latency: persistent queue with ACK retry
Phase 2-D	17/11 – 23/11/2026	Hardware Prototyping & Bench Testing	Differential chassis, RPLiDAR A1, ESP32-S3 LoRa beacons, BME680 gas sensor, servo dispenser bench test	Procured BOM	Hardware delay: modular simulation-first design
Phase 2-E	24/11 – 30/11/2026	System Integration & Pitch Deck	Full mission run, pitch presentation (5 min), user manual, code cleanup	Phase 2-C, 2-D	Integration bugs: frozen interfaces from Phase 1
Final	01/12/2026	Final Challenge Submission	Complete submission package, demo video, hardware logs	All milestones	—

3. TESTING AND VERIFICATION PLAN

- Unit Testing:** Automated testing of 16-byte beacon packing/unpacking, CRC-16 integrity, and geodetic coordinate transformations.
- Simulation Verification:** Headless Webots execution over $N = 20$ randomized trials measuring map IoU, beacon localization error, and mission completion rate.
- Fault Injection Suite:** Systematic evaluation of Writer power failure, beacon destruction, RF collision, and post-collapse route obstruction.